

LAB:-2

Containerization & Kubernetes Orchestration

Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

```
srishti2902@LAPTOP-6LVIDNI ~$ mkdir docker-demo
cd docker-demo
srishti2902@LAPTOP-6LVIDNI:~/docker-demo$ touch app.py Dockerfile
srishti2902@LAPTOP-6LVIDNI:~/docker-demo$ nano app.py
srishti2902@LAPTOP-6LVIDNI:~/docker-demo$ |
```

```
GNU nano 8.7.1 app.py *
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello! This application is running inside Docker."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-6 Copy

```
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-demo
cd docker-demo
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ touch app.py Dockerfile
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ nano app.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ nano Dockerfile
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ |
```

```
GNU nano 8.7.1 Dockerfile *
FROM python:3.11-slim

WORKDIR /app

COPY app.py .

RUN pip install flask

EXPOSE 5000

CMD ["python", "app.py"]

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line ^E Redo       ^-6 Copy
```

```
srishiti2902@LAPTOP-6LVIDNI:~$ mkdir docker-demo
cd docker-demo
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ touch app.py Dockerfile
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ nano app.py
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ nano Dockerfile
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ docker build -t docker-demo:1.0 .
[+] Building 108.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:9534e5a8e315485d4861ed659af8fd78a284c015f9b73661b41d6bab25604534
=> => resolve docker.io/library/python:3.11-slim@sha256:9534e5a8e315485d4861ed659af8fd78a284c015f9b73661b41d6bab25604534
=> => sha256:3678bb828654fcc3752f7a5eb63c5accfa0c97847252f25485285550d74a3fd 4.27MB / 4.27MB
=> => sha256:db840d886b65cf73e0feea3bd0cf11863818114e466846ae3f2b2fb1a8c2b143 14.45MB / 14.45MB
=> => sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413514de8e413b51f90e91a 249B / 249B
=> => extracting sha256:3678bb828654fcc3752f7a5eb63c5accfa0c97847252f25485285550d74a3fd
=> => extracting sha256:db840d886b65cf73e0feea3bd0cf11863818114e466846ae3f2b2fb1a8c2b143
=> => extracting sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413514de8e413b51f90e91a
=> [internal] load build context
=> => transferring context: 241B
=> [2/4] WORKDIR /app
=> [3/4] COPY app.py
=> [4/4] RUN pip install flask
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:8e2ca3b460d6f5ed6e17bff9482aac3e016c5603799b3c5a3d026fa612bbabbb
=> => exporting config sha256:da7cf2bd14b2c2749c578959a3616d7ef43407a7dc863d84040df15f7fc11dfd
=> => exporting attestation manifest sha256:6d15ade0827886590970412b13be886d1f92e25e0df0c938c9828f92a60a3e5a
=> => exporting manifest list sha256:3b67be2e461d4caa793c5481df8539775deb8346f49cc92789867c5e54790df3
=> => naming to docker.io/library/docker-demo:1.0
=> => unpacking to docker.io/library/docker-demo:1.0
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$
```

```
srishiti2902@LAPTOP-6LVIDNI:~$ mkdir docker-demo
cd docker-demo
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ touch app.py Dockerfile
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ nano app.py
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ nano Dockerfile
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ docker build -t docker-demo:1.0 .
[+] Building 108.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:9534e5a8e315485d4861ed659af8fd78a284c015f9b73661b41d6bab25604534
=> => resolve docker.io/library/python:3.11-slim@sha256:9534e5a8e315485d4861ed659af8fd78a284c015f9b73661b41d6bab25604534
=> => sha256:3678bb828654fcc3752f7a5eb63c5accfa0c97847252f25485285550d74a3fd 4.27MB / 4.27MB
=> => sha256:db840d886b65cf73e0feea3bd0cf11863818114e466846ae3f2b2fb1a8c2b143 14.45MB / 14.45MB
=> => sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413514de8e413b51f90e91a 249B / 249B
=> => extracting sha256:3678bb828654fcc3752f7a5eb63c5accfa0c97847252f25485285550d74a3fd
=> => extracting sha256:db840d886b65cf73e0feea3bd0cf11863818114e466846ae3f2b2fb1a8c2b143
=> => extracting sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413514de8e413b51f90e91a
=> [internal] load build context
=> => transferring context: 241B
=> [2/4] WORKDIR /app
=> [3/4] COPY app.py
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=> => exporting config sha256:da7cf2bd14b2c2749c578959a3616d7ef43407a7dc863d84040df15f7fc11dfd
=> => exporting attestation manifest sha256:6d15ade0827886590970412b13be886d1f92e25e0df0c938c9828f92a60a3e5a
=> => exporting manifest list sha256:3b67be2e461d4caa793c5481df8539775deb8346f49cc92789867c5e54790df3
=> => naming to docker.io/library/docker-demo:1.0
=> => unpacking to docker.io/library/docker-demo:1.0
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ docker images

Info + U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
docker-demo:1.0 3b67be2e461d 224MB 55.1MB
hello-world:latest 5dd0d3e6e255 25.9kB 9.49kB
nginx:latest b34848eff6db 241MB 66.2MB
python:3.12-slim 09f7da3bc104 179MB 45.4MB
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ docker run -d --name docker-demo-container -p 5000:5000 docker-demo:1.0
45a04f4d0e37cbae9d77ac3e9b5d46bb3780cc6d5fc93b21cfe3d4b20130cd80
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
45a04f4d0e37 docker-demo:1.0 "python app.py" 41 seconds ago Up 38 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp docker-demo-container
srishiti2902@LAPTOP-6LVIDNI:~/docker-demo$
```



Hello! This application is running inside Docker.

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker run -d --name docker-demo-container -p 5000:5000 docker-demo
45a04f4d0e37c3bae9d77ac3e9b5d46bb3780cc6d5fc93b21cfe3d4b20130cd80
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
45a04f4d0e37   docker-demo:1.0 "python app.py"         41 seconds ago Up 38 seconds  0.0.0.0:5000->5000/tcp, [
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ curl http://localhost:5000
Hello! This application is running inside Docker.srishti2902@LAPTOP-6LVIDNHT:~/docke
```

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker history docker-demo:1.0
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
3b67be2e461d   5 hours ago    CMD ["python" "app.py"] 0B        buildkit.dockerfile.v0
<missing>      5 hours ago    EXPOSE [5000/tcp]      0B        buildkit.dockerfile.v0
<missing>      5 hours ago    RUN /bin/sh -c pip install flask # buildkit 19.7MB    buildkit.dockerfile.v0
<missing>      5 hours ago    COPY app.py . # buildkit 12.3kB    buildkit.dockerfile.v0
<missing>      5 hours ago    WORKDIR /app           8.19kB    buildkit.dockerfile.v0
<missing>      6 days ago     CMD ["python3"]         0B        buildkit.dockerfile.v0
<missing>      6 days ago     RUN /bin/sh -c set -eux; for src in idle3 p... 16.4kB    buildkit.dockerfile.v0
<missing>      6 days ago     RUN /bin/sh -c set -eux; savedAptMark="$(a... 48.8MB    buildkit.dockerfile.v0
<missing>      6 days ago     ENV PYTHON_SHA256=91bcdebfdde239a003ae93738a... 0B        buildkit.dockerfile.v0
<missing>      6 days ago     ENV PYTHON_VERSION=3.11.16 0B        buildkit.dockerfile.v0
<missing>      6 days ago     ENV GPG_KEY=A035C8C19219BA821ECEA86B64E628F8... 0B        buildkit.dockerfile.v0
<missing>      6 days ago     RUN /bin/sh -c set -eux; apt-get update; a... 13.2MB    buildkit.dockerfile.v0
<missing>      6 days ago     ENV LANG=C.UTF-8        0B        buildkit.dockerfile.v0
<missing>      6 days ago     ENV PATH=/usr/local/bin:/usr/local/sbin:/usr... 0B        buildkit.dockerfile.v0
<missing>      2 weeks ago    # debian.sh --arch 'amd64' out/ 'trixie' '@1... 87.5MB    debuerreotype 0.17
```

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop docker-demo-container
docker-demo-container
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  STATUS        PORTS        NAMES
45a04f4d0e37   docker-demo:1.0 "python app.py"         Exited (255) 5 minutes ago  0.0.0.0:5000->5000/tcp  docker-demo-container
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker start docker-demo-container
docker-demo-container
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
45a04f4d0e37   docker-demo:1.0 "python app.py"         5 hours ago    Up 11 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  docker-demo-container
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$
```

2. Manage Docker containers and images

Manage Docker containers and images: list, start, stop, remove containers, tag and push images to a registry, and use Docker Compose for multi-container setups.

1. Check Docker is working

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker --version
docker compose version
Docker version 29.7.2, build a7dcaa6
Docker Compose version v5.4.0
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
45a04f4d0e37   docker-demo:1.0 "python app.py"         5 hours ago    Up 10 minutes  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  docker-demo-container
```

2. List containers

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-container
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-container

3. Start a container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker run -d --name mycontainer nginx
70f85dd876823dab8ee28efe6828c7103f41f9e99bec580e22364f0b8e861c45
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
70f85dd87682	nginx	"/docker-entrypoint..."	7 seconds ago	Up 7 seconds	80/tcp	mycontainer
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 10 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-containe

4. Stop the container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop mycontainer
mycontainer
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
70f85dd87682	nginx	"/docker-entrypoint..."	22 seconds ago	Exited (0) 8 seconds ago		mycontai
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 11 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-d

5. Start the stopped container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker start mycontainer
mycontainer
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
70f85dd87682	nginx	"/docker-entrypoint..."	About a minute ago	Up 6 seconds	80/tcp	mycontainer
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 11 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-con

6. Remove the container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker stop mycontainer
mycontainer
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker rm mycontainer
mycontainer
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04f4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 12 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-container

7. List Docker images

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	b34848eff6db	241MB	66.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	b34848eff6db	241MB	66.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	

8. Pull an image

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
f340c1b7c1d6: Pull complete
02fc02c4ab8d: Pull complete
a44b5c8be616: Pull complete
956faab5efb3: Pull complete
c12f394dea35: Pull complete
07db7bf2649b: Pull complete
76f27c02d218: Download complete
25202a7045eb: Download complete
Digest: sha256:05b8cb60c354a44ab824ea6e7dc69b46d50762cdbc728a347a5b656e6fb3d7c4
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	05b8cb60c354	253MB	69.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	

9. Tag an image

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker tag nginx:latest srishti2902/my-nginx:1.0
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	05b8cb60c354	253MB	69.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	
srishti2902/my-nginx:1.0	05b8cb60c354	253MB	69.2MB	

10. Login to Docker Hub

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: MDVN-HNSF
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
```

11. Push the image

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker push srishti2902/my-nginx:
1.0
The push refers to repository [docker.io/srishti2902/my-nginx]
6310eb16bf42: Unavailable
02fc02c4ab8d: Unavailable
956faab5efb3: Unavailable
a44b5c8be616: Unavailable
f340c1b7c1d6: Unavailable
07db7bf2649b: Unavailable
c12f394dea35: Unavailable
push access denied, repository does not exist or may require authorization: server message: insufficient_scope: authorization failed
```

12. Verify the image

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB	U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB	
nginx:latest	05b8cb60c354	253MB	69.2MB	
python:3.12-slim	09f7da3bc104	179MB	45.4MB	
srishti2902/my-nginx:1.0	05b8cb60c354	253MB	69.2MB	

13. Create a Compose project

```
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
docker-demo:1.0 3b67be2e461d 224MB 55.1MB U
hello-world:latest 5dd0d3e6e255 25.9kB 9.49kB
nginx:latest 05b8cb60c354 253MB 69.2MB
python:3.12-slim 09f7da3bc104 179MB 45.4MB
srishti2902/my-nginx:1.0 05b8cb60c354 253MB 69.2MB
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo$ mkdir compose-demo
cd compose-demo
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$ nano docker-compose.
yaml
srishti2902@LAPTOP-6LVIDNHT:~/docker-demo/compose-demo$
```

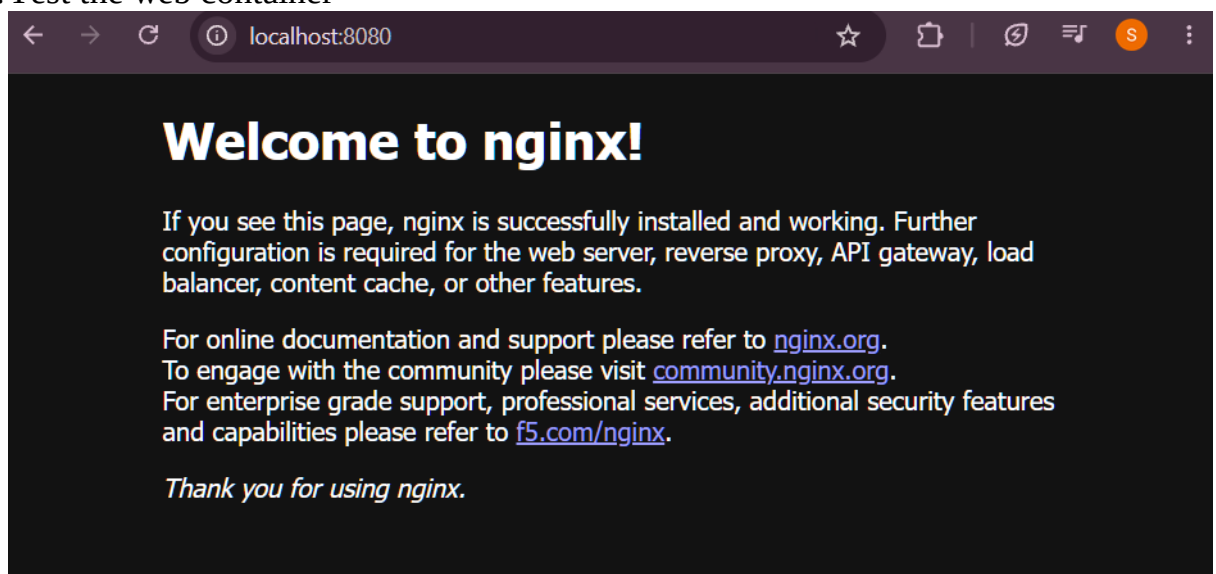
```
srishti2902@LAPTOP-6LVIDNI × + -
GNU nano 8.7.1 docker-compose.yml *
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
    depends_on:
      - redis

  redis:
    image: redis:latest
```

14. Start both containers

```
srishti2902@LAPTOP-6LVIDNI:~/docker-demo/compose-demo$ docker compose up -d
[+] up 12/12
✓Image redis:latest Pulled 9.6s
✓Network compose-demo_default Created 0.2s
✓Container compose-demo-redis-1 Started 1.5s
✓Container compose-demo-web-1 Started 1.9s
srishti2902@LAPTOP-6LVIDNI:~/docker-demo/compose-demo$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED         STATUS         PORTS
compose-demo-redis-1 redis:latest       "docker-entrypoint.s..." redis       7 seconds ago   Up 6 seconds   6379/tcp
compose-demo-web-1  nginx:latest      "/docker-entrypoint..." web         7 seconds ago   Up 5 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
```

15. Test the web container



```

srishti2902@LAPTOP-6LVINDHT:~/docker-demo/compose-demo$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>

```

16. Check the Compose logs

```

srishti2902@LAPTOP-6LVINDHT:~/docker-demo/compose-demo$ docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/09/07 14:54:40 [notice] 1#1: using the "epoll" event method
web-1 | 2026/09/07 14:54:40 [notice] 1#1: nginx/1.31.5
web-1 | 2026/09/07 14:54:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/09/07 14:54:40 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1 | 2026/09/07 14:54:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker processes
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 30
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 31
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 32
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 33
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 34
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 35
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 36
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 37
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 38
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 39
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 40
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 41
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/152.0.0.0 Safari/537.36" "-"
redis-1 | Starting Redis Server
redis-1 | 1:C 07 Sep 2026 14:54:39.929 * o080o080o080o Redis is starting o080o080o080o
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Redis version=8.10.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Configuration loaded
redis-1 | 1:M 07 Sep 2026 14:54:39.931 * monotonic clock: POSIX clock_gettime
redis-1 | 1:M 07 Sep 2026 14:54:39.937 * Running mode=standalone, port=6379.
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> RedisBloom version 8.10.1 (Git=unknown)
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-error-rate : 0.01 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-initial-size : 100 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-expansion-factor : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-bucket-size : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-initial-size : 1024 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-iterations : 20 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-expansion-factor : 1 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-expansions : 32 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> ]
redis-1 | 1:M 07 Sep 2026 14:54:39.939 * Module 'bf' loaded from /usr/local/lib/redis/modules/redisbloom.so
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> search-workers default: 12 (min of MAX_WORKER_THREADS=16 and CPU cores)
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> Redis version found by RedisSearch : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> RedisSearch version 8.10.0 (Git=)
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms):
500, timeout policy: return, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search res
ults: 1000000, default scorer: BM25STD,
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Initialized thread pools!
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled workers threadpool of size 12
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to config changes
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster slot migration events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster topology change events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled role change notification
redis-1 | 1:M 07 Sep 2026 14:54:39.976 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
redis-1 | 1:M 07 Sep 2026 14:54:39.978 * Module 'search' loaded from /usr/local/lib/redis/modules/redisearch.so
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> RedisTimeSeries version 81000, git_sha=
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Redis version found by RedisTimeSeries : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-compaction-policy : }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-num-threads : 3 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-libm-protocol : INTERNAL }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-retention-policy : 0 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-duplicate-policy : block }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-chunk-size-bytes : 4096 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-encoding : compressed }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> { ts-ignore-max-time-diff: 0 }

```

```

srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose logs web
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/09/07 14:54:40 [notice] 1#1: using the "epoll" event method
web-1 | 2026/09/07 14:54:40 [notice] 1#1: nginx/1.31.5
web-1 | 2026/09/07 14:54:40 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/09/07 14:54:40 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1 | 2026/09/07 14:54:40 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker processes
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 30
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 31
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 32
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 33
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 34
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 35
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 36
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 37
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 38
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 39
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 40
web-1 | 2026/09/07 14:54:40 [notice] 1#1: start worker process 41
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"
web-1 | 2026/09/07 14:55:11 [error] 31#31: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.18.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:11 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"
web-1 | 172.18.0.1 - - [07/Sep/2026:14:55:27 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"

```

```

srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose logs redis
redis-1 | Starting Redis Server
redis-1 | 1:C 07 Sep 2026 14:54:39.929 * o000o000o000 Redis is starting o000o000o000
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Redis version=8.10.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1 | 1:C 07 Sep 2026 14:54:39.930 * Configuration loaded
redis-1 | 1:M 07 Sep 2026 14:54:39.931 * monotonic clock: POSIX clock_gettime
redis-1 | 1:M 07 Sep 2026 14:54:39.937 * Running mode=standalone, port=6379.
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> RedisBloom version 8.10.1 (Git=unknown)
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-error-rate : 0.01 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-initial-size : 100 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { bf-expansion-factor : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-bucket-size : 2 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-initial-size : 1024 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-iterations : 20 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-expansion-factor : 1 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> { cf-max-expansions : 32 }
redis-1 | 1:M 07 Sep 2026 14:54:39.938 * <bf> ]
redis-1 | 1:M 07 Sep 2026 14:54:39.939 * Module 'bf' loaded from /usr/local/lib/redis/modules/redisbloom.so
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> search-workers default: 12 (min of MAX_WORKER_THREADS=16 and CPU cores)
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> Redis version found by RedisSearch : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.974 * <search> RedisSearch version 8.10.0 (Git=)
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy: return, oom policy: return, cursor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search res
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Initializd thread pools!
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled workers threadpool of size 12
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to config changes
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster slot migration events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Subscribe to cluster topology change events
redis-1 | 1:M 07 Sep 2026 14:54:39.975 * <search> Enabled role change notification
redis-1 | 1:M 07 Sep 2026 14:54:39.976 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
redis-1 | 1:M 07 Sep 2026 14:54:39.978 * Module 'search' loaded from /usr/local/lib/redis/modules/redisearch.so
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> RedisTimeSeries version 81000, git_sha=
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Redis version found by RedisTimeSeries : 8.10.1 - oss
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> Registering configuration options: [
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-compaction-policy : 3 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-num-threads : 3 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-libmtr-protocol : INTERNAL }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-retention-policy : 0 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-duplicate-policy : block }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-chunk-size-bytes : 4096 }
redis-1 | 1:M 07 Sep 2026 14:54:39.979 * <timeseries> { ts-encoding : compressed }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> { ts-ignore-max-time-diff : 0 }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> { ts-ignore-max-val-diff : 0.000000 }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> { ts-topology-events : true }
redis-1 | 1:M 07 Sep 2026 14:54:39.980 * <timeseries> ]

```

17. Stop the Compose setup

```

srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose stop
[+] stop 2/2
✓Container compose-demo-web-1 Stopped 0.6s
✓Container compose-demo-redis-1 Stopped 0.3s
srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
compose-demo-redis-1	redis:latest	"docker-entrypoint.s..."	redis	2 minutes ago	Up 5 seconds	6379/tcp
compose-demo-web-1	nginx:latest	"/docker-entrypoint.s..."	web	2 minutes ago	Up 5 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

18. Start the Compose setup again

```

srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose start
[+] start 2/2
✓Container compose-demo-redis-1 Started 0.5s
✓Container compose-demo-web-1 Started 0.5s
srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
compose-demo-redis-1	redis:latest	"docker-entrypoint.s..."	redis	2 minutes ago	Up 5 seconds	6379/tcp
compose-demo-web-1	nginx:latest	"/docker-entrypoint.s..."	web	2 minutes ago	Up 5 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

19. Remove the Compose containers

```

srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker compose down
[+] down 3/3
✓Container compose-demo-web-1 Removed 0.4s
✓Container compose-demo-redis-1 Removed 0.4s
✓Network compose-demo_default Removed 0.3s
srishti2902@LAPTOP-6LVINDNH:~/docker-demo/compose-demo$ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
45a04fd4d0e37	docker-demo:1.0	"python app.py"	5 hours ago	Up 21 minutes	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	docker-demo-container

3. Docker Storage:-

1. Loss of container-layer data after container deletion.
2. Permanent storage using a Docker named volume and bind mount

1. Create a container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name storage-test nginx
8f65d13fd5b8563d922247ccf18faceca47d05f847e3a6c20c632ebacdfd231e
srishti2902@LAPTOP-6LVIDNHT:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
8f65d13fd5b8	nginx	"/docker-entrypoint..."	11 seconds ago	Up 10 seconds	80/tcp	storage-test

2. Create data inside the container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec storage-test sh -c 'echo 'This data is stored in the container layer' > /tmp/test.txt'
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec storage-test cat /tmp/test.txt
This data is stored in the container layer
```

3. Delete the container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker stop storage-test
storage-test
srishti2902@LAPTOP-6LVIDNHT:~$ docker rm storage-test
storage-test
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name storage-test nginx
3c26a71ec5650d2c19205eb712a289149a5228d90da37f55a62fbda4c6a70a59
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec storage-test cat /tmp/test.txt
cat: /tmp/test.txt: No such file or directory
```

4. Create a new container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name storage-test nginx
3c26a71ec5650d2c19205eb712a289149a5228d90da37f55a62fbda4c6a70a59
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec storage-test cat /tmp/test.txt
cat: /tmp/test.txt: No such file or directory
```

5. Create a named volume

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker volume create myvolume
myvolume
srishti2902@LAPTOP-6LVIDNHT:~$ docker volume ls
```

DRIVER	VOLUME NAME
local	myvolume

6. Attach the volume to a container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker volume create myvolume
myvolume
srishti2902@LAPTOP-6LVIDNHT:~$ docker volume ls
```

DRIVER	VOLUME NAME
local	myvolume

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name volume-test -v myvolume:/data nginx
42c818f56b83369de8bc172da798c81df70304ddcf8a5cd9438deab18c5a4baa
```

7. Store data in the volume

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec volume-test sh -c 'echo 'This data is stored permanently in a Docker volume' > /data/test.txt'
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec volume-test cat /data/test.txt
This data is stored permanently in a Docker volume
```

8. Delete the container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker stop volume-test
docker rm volume-test
volume-test
volume-test
```

9. Create another container using the same volume

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name volume-test-2 -v myvolume:/data nginx
c61b7cc1451807edfd5d8d2a57af30580de055954027a4da913838a4c7e4a4f51
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec volume-test-2 cat /data/test.txt
This data is stored permanently in a Docker volume
```

10. Create a directory on your computer

```
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir ~/docker-data
srishti2902@LAPTOP-6LVIDNHT:~$ ls ~/docker-data
```

11. Create a container with a bind mount

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name bind-test -v ~/docker-data:/data nginx
a15627623259cee699ac0c3f6beb427a2aef51afddc26f11d7f6fb99a57351dc
```

12. Create data inside the container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec bind-test sh -c "echo 'This data is stored using a bind mount' > /data/test.txt"
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec bind-test cat /data/test.txt
This data is stored using a bind mount
srishti2902@LAPTOP-6LVIDNHT:~$ cat ~/docker-data/test.txt
This data is stored using a bind mount
```

13. Delete the container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker stop bind-test
bind-test
srishti2902@LAPTOP-6LVIDNHT:~$ docker rm bind-test
bind-test
srishti2902@LAPTOP-6LVIDNHT:~$ cat ~/docker-data/test.txt
This data is stored using a bind mount
srishti2902@LAPTOP-6LVIDNHT:~$
```

4. Docker Network:-

- Communication between two containers through a Docker network.
- Communication using container names and container IP addresses.

1. Create a Docker network

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker network create mynetwork
ef17bab37cebbface4d04bbe978ec896416512999a475ee2fb1afe29fd5f9cf9
srishti2902@LAPTOP-6LVIDNHT:~$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
135790a3248d        bridge              bridge              local
c3bbb4b3f17d        host                host                local
ef17bab37ceb        mynetwork           bridge              local
11e0b59157cb        none                null                local
```

2. Create the first container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name container1 --network mynetwork ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
f3db1cd94078: Pull complete
06e9d71331fb: Pull complete
f79afc4bbefe: Download complete
Digest: sha256:2260313b31c8c011cd2eebe728008efac1b3982be73eb71348ea2648d2c0e09b
Status: Downloaded newer image for ubuntu:latest
4a46e330eac6ca107de22f7dfd9a90364ccb468a7c0db9754d04f728699a0d5
```

3. Create the second container

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name container1 --network mynetwork ubuntu
docker: Error response from daemon: Conflict. The container name "/container1" is already in use by container "4a46e330eacf6ca107de22f7df9a90364ccb468a7c0d
b9754d04f72699a0d5". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
srishti2902@LAPTOP-6LVIDNHT:~$ docker run -dit --name container2 --network mynetwork ubuntu
5a91200994f3ee1eb8a32874ab1d42d76e0d5f43223197ea0055254371ca585
srishti2902@LAPTOP-6LVIDNHT:~$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
5a91200994f3	ubuntu	"/bin/bash"	9 seconds ago	Up 9 seconds		container2
4a46e330eacf	ubuntu	"/bin/bash"	36 seconds ago	Up 35 seconds		container1
c61b7cc14518	nginx	"/docker-entrypoint..."	29 minutes ago	Up 29 minutes	80/tcp	volume-test-2

4. Install ping in both containers

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container1 apt-get update
docker exec container1 apt-get install -y iputils-ping
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [578 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [489 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [212 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [11.2 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [743 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [14.6 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [505 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [338 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3306 B]
Fetched 26.0 MB in 9s (2820 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following additional packages will be installed:
  libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
The following NEW packages will be installed:
  iputils-ping libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
0 upgraded, 5 newly installed, 0 to remove and 24 not upgraded.
Need to get 761 kB of archives.
After this operation, 2677 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcap2 amd64 1:2.75-10ubuntu2 [29.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunistring5 amd64 1.3-2build1 [610 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libidn2-0 amd64 2.3.8-4build1 [67.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 iputils-ping amd64 3:20250605-1ubuntu1 [47.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 linux-sysctl-defaults all 4.15ubuntu5 [6208 B]
debconf: unable to initialize frontend: Dialog
debconf: (TERM is not set, so the dialog frontend is not usable.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
```

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container2 apt-get update
docker exec container2 apt-get install -y iputils-ping
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [20.1 MB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [11.2 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [212 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [489 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [578 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [189 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [352 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1874 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [338 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [14.6 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [743 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [505 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3306 B]
Fetched 26.0 MB in 7s (3847 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following additional packages will be installed:
  libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
The following NEW packages will be installed:
  iputils-ping libcap2 libidn2-0 libunistring5 linux-sysctl-defaults
0 upgraded, 5 newly installed, 0 to remove and 24 not upgraded.
Need to get 761 kB of archives.
After this operation, 2677 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcap2 amd64 1:2.75-10ubuntu2 [29.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunistring5 amd64 1.3-2build1 [610 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libidn2-0 amd64 2.3.8-4build1 [67.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 iputils-ping amd64 3:20250605-1ubuntu1 [47.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 linux-sysctl-defaults all 4.15ubuntu5 [6208 B]
debconf: unable to initialize frontend: Dialog
debconf: (TERM is not set, so the dialog frontend is not usable.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/Readline.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-l
inux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-
linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 5.)
debconf: falling back to frontend: Teletype
debconf: unable to initialize frontend: Teletype
```

5. Communication using container name

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container1 ping -c 4 container2
PING container2 (172.18.0.3) 56(84) bytes of data.
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=1 ttl=64 time=3.76 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=2 ttl=64 time=0.312 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=3 ttl=64 time=0.188 ms
64 bytes from container2.mynetwork (172.18.0.3): icmp_seq=4 ttl=64 time=0.123 ms

--- container2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.123/1.094/3.755/1.537 ms
```

6. Find the IP address of container2

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' container2
172.18.0.3
```

7. Communication using container IP address

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' container2
172.18.0.3
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container1 ping -c 4 172.18.0.3
PING 172.18.0.3 (172.18.0.3) 56(84) bytes of data.
64 bytes from 172.18.0.3: icmp_seq=1 ttl=64 time=0.155 ms
64 bytes from 172.18.0.3: icmp_seq=2 ttl=64 time=0.098 ms
64 bytes from 172.18.0.3: icmp_seq=3 ttl=64 time=0.096 ms
64 bytes from 172.18.0.3: icmp_seq=4 ttl=64 time=0.120 ms

--- 172.18.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3074ms
rtt min/avg/max/mdev = 0.096/0.117/0.155/0.023 ms
```

8. Check the network details

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker network inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "ef17bab37cebbface4d04bbe978ec896416512999a475ee2fb1afe29fd5f9cf9",
    "Created": "2026-09-08T14:06:48.067539661Z",
    "Scope": "Local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {
      "com.docker.network.enable_ipv4": "true",
      "com.docker.network.enable_ipv6": "false"
    },
    "Labels": {},
    "Containers": {
      "4a46e330eacf6ca107de22f7dfd9a90364ccb468a7c0db9754d04f728699a0d5": {
        "Name": "container1",
        "EndpointID": "108608831e3503fd8a828a73fc0e54876bd6716881cb0fd24be194d9533eed9b",
        "MacAddress": "e6:1b:70:85:fe:2a",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      },
      "5a91200994f3ee1eb8a32874ab1df2cd76e0d5f43223197ea0055254371ca585": {
        "Name": "container2",
```

9. Test communication in the opposite direction

```

srishti2902@LAPTOP-6LVIDNHT:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}' container1
172.18.0.2
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 container1
PING container1 (172.18.0.2) 56(84) bytes of data.
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=1 ttl=64 time=0.243 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=2 ttl=64 time=0.485 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=3 ttl=64 time=0.144 ms
64 bytes from container1.mynetwork (172.18.0.2): icmp_seq=4 ttl=64 time=0.086 ms

--- container1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3046ms
rtt min/avg/max/mdev = 0.086/0.239/0.485/0.152 ms
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 <container1-IP>
-bash: syntax error near unexpected token `newline'
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 <container1-172.18.0.2>
-bash: syntax error near unexpected token `newline'
srishti2902@LAPTOP-6LVIDNHT:~$ docker exec container2 ping -c 4 172.18.0.2
PING 172.18.0.2 (172.18.0.2) 56(84) bytes of data.
64 bytes from 172.18.0.2: icmp_seq=1 ttl=64 time=0.212 ms
64 bytes from 172.18.0.2: icmp_seq=2 ttl=64 time=0.088 ms
64 bytes from 172.18.0.2: icmp_seq=3 ttl=64 time=0.100 ms
64 bytes from 172.18.0.2: icmp_seq=4 ttl=64 time=0.086 ms

--- 172.18.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.086/0.121/0.212/0.052 ms

```

10. Final cleanup

```

srishti2902@LAPTOP-6LVIDNHT:~$ docker stop container1 container2
container1
container2
srishti2902@LAPTOP-6LVIDNHT:~$ docker rm container1 container2
container1
container2
srishti2902@LAPTOP-6LVIDNHT:~$ docker network rm mynetwork
mynetwork
srishti2902@LAPTOP-6LVIDNHT:~$ |

```